Name:_____ Period:_____
Earth and Space Science

IGNEOUS ROCKS AND BOWEN'S REACTION SERIES LEARNING GUIDE

Learning Intention
Today we will be learning how magma forms and how its cooling rate and mineral composition shape igneous rocks so we can explain why Yosemite's granite has large crystals while Hawai‘i's basalt and obsidian have tiny crystals or none at all.

Success Criteria
I can… ● Explain how magma forms, including how water lowers the melting temperature of rock. ● Use crystal size and texture to decide whether an igneous rock formed underground (intrusive) or at the surface (extrusive). ● Use the ESSRT to connect an igneous rock's minerals to its color, density, and the order its minerals crystallized.

Yosemite: Mountains Made of Cooled Magma

Yosemite National Park, in California's Sierra Nevada mountains, is famous for its giant granite cliffs. El Capitan rises about 3,000 feet above the valley floor, and Half Dome is one of the most recognized rock formations in the world. This granite began as magma miles below the surface about 100 million years ago. It cooled so slowly that its crystals grew large enough to see without a hand lens. Millions of years of uplift and erosion, including carving by glaciers, later exposed it at the surface. Compare that to the basalt you saw in Hawai‘i, which cooled from lava in hours or days. In this guide, Yosemite and Hawai‘i are our examples of how igneous rocks form. As you read, keep this question in mind: *Why do some igneous rocks have large crystals while others have tiny crystals or none at all?*

How Magma Forms

Igneous rocks form when melted rock cools and hardens. Melted rock below Earth's surface is **magma**. Melted rock that reaches the surface is **lava**. Heat from Earth's interior makes magma form deep underground in three main ways:

- **Increasing temperature:** Rock gets hot enough to partly melt.
- **Decreasing pressure:** Hot rock deep in the mantle stays solid because of the great pressure on it. If that rock rises and the pressure drops, it can melt without getting any hotter. This is how Hawai‘i's magma forms.
- **Adding water:** Water lowers the melting temperature of rock, the same way salt lowers the melting temperature of ice. Where an ocean plate sinks into the mantle, it carries water down with it, and the rock above it begins to melt. The magma that formed Yosemite's granite came from this process.

Because magma is less dense than the solid rock around it, it rises. Some magma erupts at the surface, and some cools and hardens underground.

Check for Understanding

_____ 1. Adding water to hot rock deep underground makes the rock
(1) melt at a lower temperature
(2) melt at a higher temperature
(3) become more dense
(4) cool faster

2. What is the difference between magma and lava?

_____

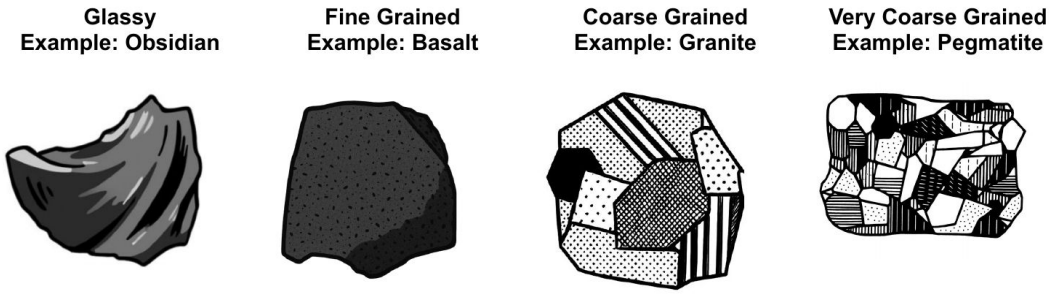
Cooling Rate, Crystal Size, and Texture

An igneous rock's **texture** describes the size of its crystals. Crystal size depends on how fast the magma or lava cooled. When melted rock cools slowly, atoms have time to move into place and build large crystals. When it cools quickly, many tiny crystals form at once, or none form at all.

- **Intrusive** igneous rocks form when magma cools **slowly underground**. They are **coarse-grained**, with crystals large enough to see. Examples: granite (Yosemite) and gabbro. Magma that cools extremely slowly can form very coarse-grained rock called pegmatite.
- **Extrusive** igneous rocks form when lava cools **quickly at the surface**. They are **fine-grained**, with crystals too small to see. Examples: basalt (Hawai‘i) and rhyolite.
- **Glassy** rocks, like obsidian, form when lava cools so fast that no crystals form.
- **Vesicular** rocks, like pumice and scoria, have holes left by gas bubbles trapped as lava cooled.
- **Porphyritic** rocks have large crystals surrounded by tiny ones. The large crystals grew while magma cooled slowly underground, and the tiny ones formed after it rose and cooled quickly.

Check for Understanding: Model It

The diagram represents four igneous rocks with different textures.



_____ 1. Which rock in the diagram formed from lava that cooled the fastest?
(1) obsidian
(2) basalt
(3) granite
(4) pegmatite

2. Write **I** if the rock is intrusive or **E** if it is extrusive.

_____ a. Granite

_____ b. Basalt

_____ c. Obsidian

_____ d. Gabbro

3. Explain why the crystals in Yosemite's granite are larger than the crystals in Hawai‘i's basalt.

_____

_____

Composition: Felsic and Mafic

Igneous rocks are also classified by the minerals they contain. **Felsic** rocks are rich in silica and made mostly of light-colored minerals, such as quartz and potassium feldspar. They are light in color and **less dense**. Granite, the rock of Yosemite and the continental crust, is felsic. **Mafic** rocks are rich in iron and magnesium and made mostly of dark minerals, such as pyroxene and olivine. They are dark in color and **more dense**. Basalt, the rock of Hawai‘i and the oceanic crust, is mafic. **Intermediate** rocks, such as diorite and andesite, fall in between, and **ultramafic** rocks, such as peridotite, come from the mantle.

Rocks with the same composition can have different textures. Granite and rhyolite have the same minerals, but granite cooled slowly underground and rhyolite cooled quickly at the surface. Gabbro and basalt are another matching pair.

Check for Understanding

_____ 1. Compared to basalt, granite is
(1) darker and more dense
(2) darker and less dense
(3) lighter and more dense
(4) lighter and less dense

2. Name the extrusive rock that has the same mineral composition as granite.

_____

Bowen's Reaction Series

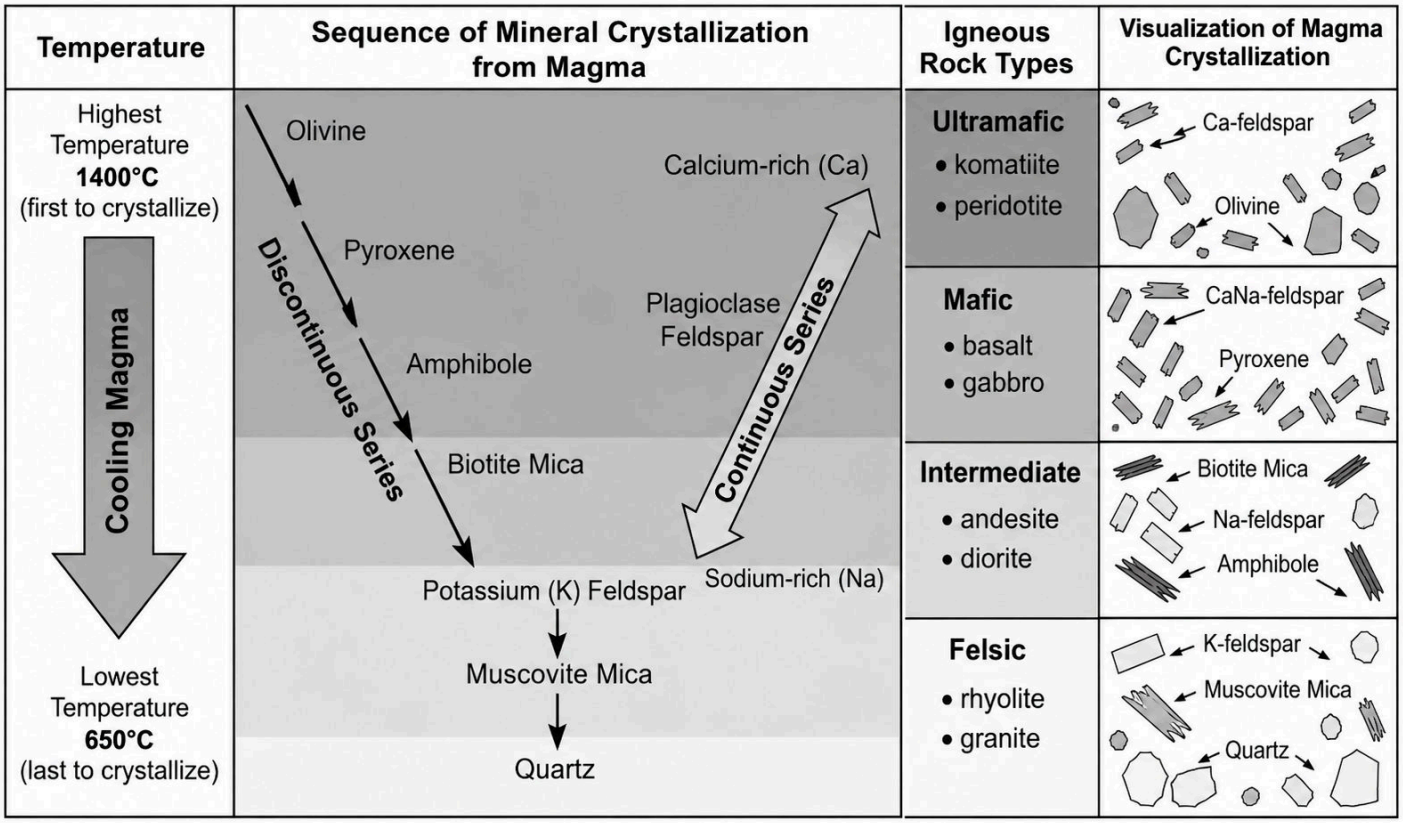
As magma cools, its minerals do not all crystallize at once. **Bowen's Reaction Series** shows the order in which minerals crystallize as magma cools from about 1,400°C to about 650°C. Olivine and pyroxene crystallize first, at the highest temperatures. Next come amphibole and biotite mica. At the same time, plagioclase feldspar crystallizes, starting out calcium-rich and becoming more sodium-rich as the magma cools. Potassium feldspar, muscovite mica, and quartz crystallize last.

This order explains the two rock families. The early, high-temperature minerals make up mafic and ultramafic rocks, and the late, low-temperature minerals make up felsic rocks like granite.

ESSRT CONNECTION

Use the Model of Bowen's Reaction Series and the Mineral Composition of Igneous Rocks chart (page 14) in the 2026 Revised Edition Reference Tables for Earth and Space Sciences.

Model of Bowen's Reaction Series

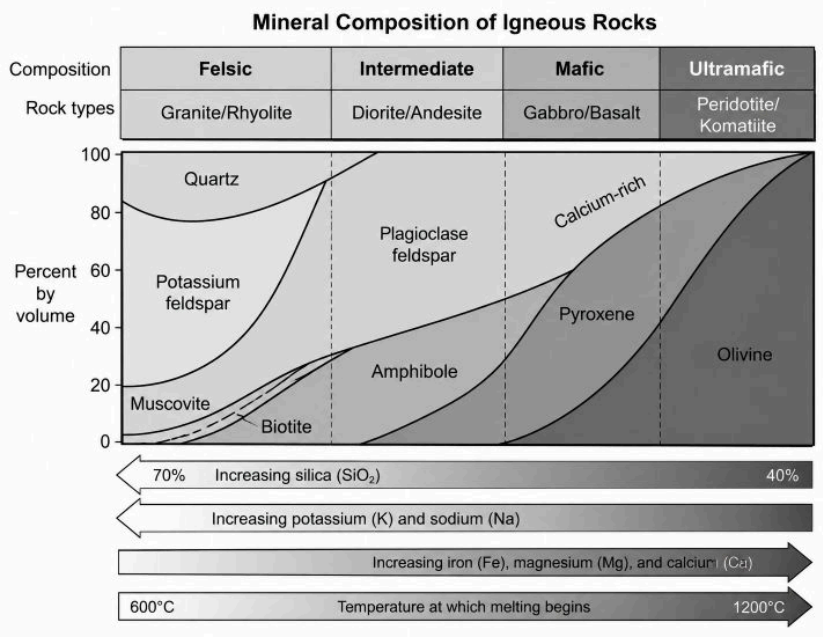


1. According to the Model of Bowen's Reaction Series, which mineral is the first to crystallize from cooling magma?

_____

_____ 2. As magma cools, plagioclase feldspar becomes
(1) more calcium-rich
(2) more sodium-rich
(3) more iron-rich
(4) more quartz-rich

_____ 3. The last minerals to crystallize from magma are found mainly in which igneous rocks?
(1) komatiite and peridotite
(2) basalt and gabbro
(3) andesite and diorite
(4) rhyolite and granite



_____ 4. According to the Mineral Composition of Igneous Rocks chart, compared to felsic rocks, mafic rocks contain more

(1) quartz
(2) potassium feldspar
(3) muscovite
(4) pyroxene

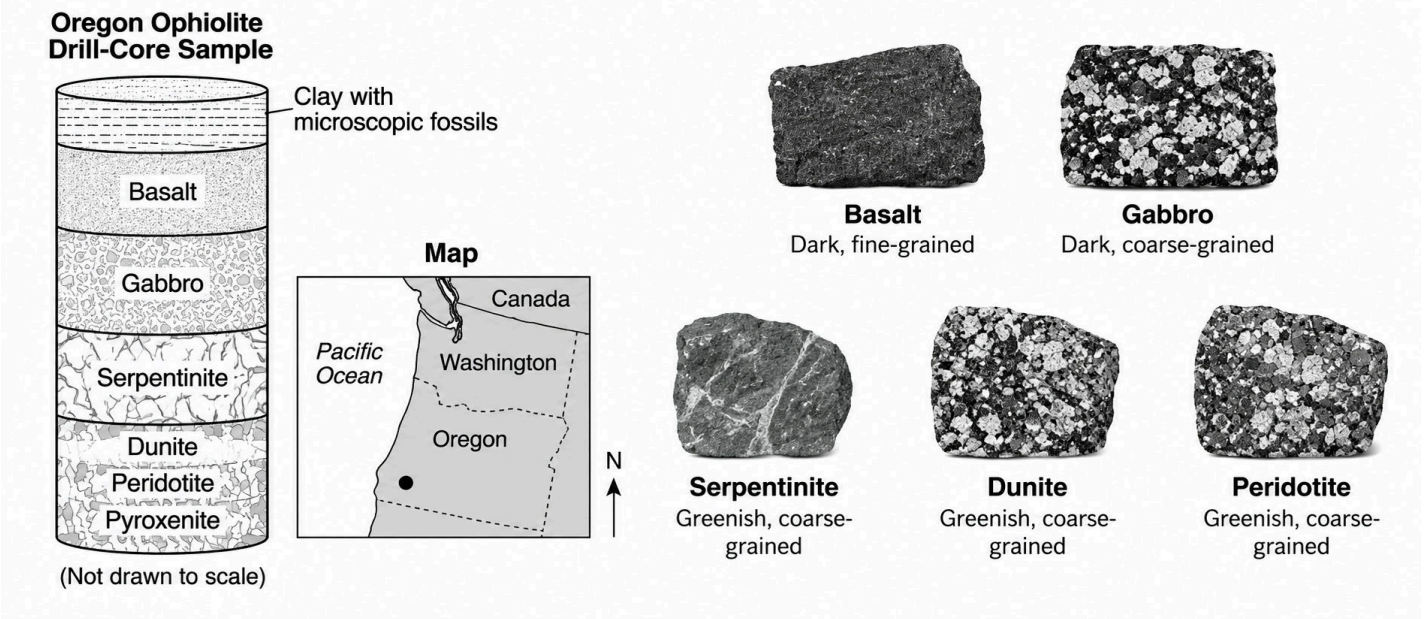
5. According to the chart, the temperature at which melting begins increases from about _____ for felsic rocks to about _____ for ultramafic rocks.

Cluster Set 1

Base your answers to questions 1 through 4 on the passage and diagram below and on your knowledge of Earth and Space Science.

Ophiolites: Oceanic Crust on Land

An ophiolite is a slice of oceanic crust and upper mantle that has been pushed up onto land. Ophiolites let scientists study rocks that formed beneath the ocean floor at different depths. The diagram shows a drill-core sample from an ophiolite in Oregon, along with the texture of each rock.



_____ 1. Which igneous rock in the drill core most likely formed closest to Earth's surface?
(1) peridotite
(2) dunite
(3) gabbro
(4) basalt

_____ 2. Which evidence best supports the claim that basalt cooled faster than gabbro?
(1) Basalt is fine-grained, while gabbro is coarse-grained.
(2) Basalt is found above gabbro in the drill core.
(3) Basalt is darker in color than gabbro.
(4) Basalt is found in Oregon.

_____ 3. Gabbro and basalt have the same mineral composition but different textures because they
(1) formed from different types of magma
(2) cooled at different rates in different locations
(3) are different types of rock (one igneous, one sedimentary)
(4) formed at different times of year

4. Peridotite is found at the bottom of the drill core and is coarse-grained. Use this evidence to describe where and how peridotite formed.

_____

_____

Mini Glossary

Bowen's Reaction Series – a model of the order in which minerals crystallize as magma cools

Coarse-grained – having crystals large enough to see without a hand lens

Extrusive – formed from lava that cooled quickly at or near Earth's surface

Felsic – rich in silica and light-colored minerals; light in color and less dense (example: granite)

Fine-grained – having crystals too small to see without magnification

Glassy – having no crystals because the lava cooled extremely fast (example: obsidian)

Igneous rock – rock that forms when magma or lava cools and hardens

Intrusive – formed from magma that cooled slowly underground

Lava – melted rock that has reached Earth's surface

Mafic – rich in iron and magnesium and dark minerals; dark in color and more dense (example: basalt)

Magma – melted rock below Earth's surface

Ophiolite – a slice of oceanic crust and upper mantle pushed up onto land

Porphyritic – having large crystals surrounded by tiny crystals, from two cooling rates

Texture – the size of the crystals in an igneous rock

Ultramafic – very rich in iron and magnesium; comes from the mantle (example: peridotite)

Vesicular – having holes left by gas bubbles (examples: pumice, scoria)